

Burak Bekir ÇAKIRER

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Summary

Electrical and Electronics Engineering student focused on embedded hardware, power electronics, PCB design, and signal integrity. Hands-on experience with STM32-based firmware, peripheral integration, real-time sensor acquisition, power-converter design, oscilloscope-based validation, and EMI/EMC-oriented hardware development. Interested in reliable embedded systems, electric-vehicle electronics, autonomous platforms, and defense-oriented hardware systems.

Education

Abdullah Gül University

2022 – Present

B.Sc. in Electrical and Electronics Engineering

Kayseri, Turkey

GPA: 3.08 / 4.00

Relevant Coursework: Embedded Systems, Signals and Systems, Electronics, Control Systems, Digital Design, Electromagnetics

Experience

Hardware Design Engineering Intern

Jun 2026 – Jul 2026

Vestel Elektronik A.Ş. – Electric Vehicle Charger (EVC) Hardware Design Team

Manisa, Turkey

- Developed a modular SH1107 OLED driver in C for the STM32G431RB microcontroller using HAL, Fast-Mode I²C at 400 kHz, a 2048-byte framebuffer, and integer-based Bresenham graphics routines.
- Interfaced the SH1107 display using an ATmega328P-based Arduino Pro Mini and a Raspberry Pi Zero W, comparing memory-constrained embedded C execution with Linux-based I²C driver layers.
- Integrated the ICM-20601 six-axis IMU over I²C and full-duplex SPI; diagnosed SDA bus lockup and restored communication by optimizing pull-up resistance and signal timing.
- Designed and analyzed power-management blocks including MP2386 and TPS562201 buck converters, an AP2111 LDO stage, a charge-pump stage, and a 3.3 V-to-5 V PWM level shifter.
- Performed oscilloscope-based validation of converter ripple, bootstrap behavior, PWM fan control, and tachometer feedback; used a near-field probe to investigate EMI emissions.
- Participated in EVC panel and Power Distribution Board replacement activities and reviewed EMC testing setups including anechoic chamber, ESD, EFT/Burst, and Surge tests.
- Analyzed high-speed PCB design practices including controlled impedance, differential routing, crosstalk reduction, length matching, stack-up constraints, and creepage/clearance requirements.

Embedded Systems Engineering Intern

Jun 2025 – Aug 2025

MTM Biotech

Turkey

- Developed BMI323 IMU firmware on STM32, Arduino, and nRF52 platforms using register-level I²C and SPI communication.
- Implemented interrupt-driven motion detection, BLE-based live data transmission, and SPI SD-card logging with buffered CSV export.
- Debugged hardware communication and pin-conflict issues, improving sensor stability and repeatability across multiple devices.
- Documented the system architecture, firmware workflow, test procedures, and debugging process.

Projects

Autonomous Defense System Power Board Design

Ongoing

TEKNOFEST Çelikkubbe Air Defense Systems Competition

- Designing a custom multi-rail Power Distribution Board in Altium Designer for motors, microcontrollers, sensors, and laser subsystems, with separated high-power and low-power domains.
- Incorporating emergency-stop, controlled switching, regulation, protection, grounding, thermal, and current-carrying requirements for reliable operation.

Autonomous Guided Vehicle (AGV)

Ongoing

Capstone Project – Abdullah Gül University

- Developing an autonomous mobile robot using Raspberry Pi for high-level processing and Arduino for low-level motor control.
- Implementing A* global planning, Dynamic Window Approach, PID speed control, and LiDAR–odometry integration for navigation and obstacle avoidance.

Touch Sensor Interface & Analog Circuit Design

Jan 2026

Abdullah Gül University

- Designed and experimentally validated an analog readout circuit that converts small capacitance changes into a repeatable DC voltage output.
- Implemented AC excitation, a fully discrete transistor-level amplifier, and AC–DC conversion; verified the complete system through LTspice and laboratory measurements.

Switchable Counter & Variable Pulse Generator

Apr 2025

Digital Systems Design Capsule Project – Abdullah Gül University

- Designed a switchable BCD counter and adjustable-duty-cycle pulse generator using astable multivibrators, a CMOS 2x2 multiplexer, and D flip-flop logic.
- Produced a 0–50% duty-cycle PWM output for LED and motor-control applications and verified operation in LTspice and on a physical breadboard.

Digital Stethoscope System

Jan 2025

SAPA Project – Abdullah Gül University

- Built and tested a digital stethoscope with TL072-based analog filtering for heart and lung sounds and Arduino-based data acquisition.
- Developed a MATLAB GUI for real-time visualization and FFT analysis, integrating analog hardware, embedded acquisition, and signal processing.

Technical Skills

Programming: C, C++, Python, MATLAB

Microcontrollers / SBCs: STM32G431RB (ARM Cortex-M4), ATmega328P (Arduino Pro Mini), nRF52, Raspberry Pi Zero W

Embedded Interfaces: I²C, SPI, UART, BLE, PWM, ADC, Timers, Interrupts

Power and Hardware: Buck Converters, LDOs, MOSFET Gate Driving, Analog/Digital Circuits, Op-Amps, BJTs, Filters

PCB and Signal Integrity: Altium Designer, Controlled Impedance, Differential Pairs, Length Matching, Crosstalk, Creepage/Clearance, EMI/EMC

Tools: STM32CubeIDE, LTspice, Oscilloscope, Signal Generator, Near-Field Probe, Git

Languages: Turkish (Native), English (Upper-Intermediate)